Yi Yang

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Professional Summary

Third-year Computer Science Ph.D. researcher developing fragment-aware molecular representations and generative AI for 3D molecule generation, alongside memory-augmented LLM systems for structured information extraction and multi-session reasoning. Combines molecular ML research with pharmaceutical R&D experience in mRNA therapeutics.

Education

University of Texas at Dallas — Dallas, TX
Ph.D. in Computer Science, GPA: 3.8 — *Expected Dec. 2028*
– Erik Jonsson Graduate Scholarship in Computer Science, 2023

Ocean University of China — Qingdao, China
M.S. in Pharmaceutical Engineering — *2018*

Technical Skills

Scientific ML: PyTorch, PyTorch Geometric, RDKit, GNNs, Transformers, graph-BPE, SE(3) modeling, XGBoost, SHAP
LLM Systems: RAG, graph-structured memory, knowledge graphs, vLLM, Pydantic, structured outputs, LLM-as-a-judge
Programming & ML Engineering: Python, SQL, FastAPI, Docker, MLflow, Slurm/HPC, Git, scikit-learn, pandas, NumPy

Research Experience

Graduate Research Assistant | University of Texas at Dallas — Aug. 2024 – Present

PoseWords: SE(3)-Aware Generative Modeling for 3D Molecules — *May 2026 – Present*

– Designing and implementing an SE(3)-aware molecular representation that factorizes 3D structures into graph-BPE fragment identity, internal conformation, and relative-pose tokens using RDKit and PyTorch Geometric.

– Developing a constrained autoregressive Transformer for peptide- and property-conditioned generation, with evaluation protocols for reconstruction RMSD, chemical validity, pose recovery, compression, and throughput.

BiScale-GTR: Fragment-Aware Graph Transformer [Paper Link] — *Oct. 2024 – May 2026*

– Developed a self-supervised PyTorch/PyG architecture combining GINE atom encoders, attention pooling, gated atom–fragment fusion, and a structure-aware Transformer for molecular property prediction.

– Built a deterministic RDKit graph-BPE pipeline with WL hashing, chemical-validity filtering, and recursive OOV decomposition; improved Peptides-func AP from 0.632 to 0.672 and Peptides-struct MAE from 0.293 to 0.262 using 78.5% less pretraining data than 2M-molecule baselines.

Interpretable ML for Organic Photovoltaic Property Prediction [Paper Link] — *Sep. 2024 – Aug. 2025*

– Built an end-to-end explainable XGBoost pipeline to curate molecular data, compute 2,000+ descriptors, train regression models, and predict photovoltaic performance across 350 donors and six acceptors, achieving test $R^2 = 0.918$.

– Performed feature selection, SHAP-based attribution and performance analysis, and external-dataset validation; identified 10 interpretable predictors and quantified model transferability beyond the training distribution.

Industry Experience

Team Lead, StemiRNA Therapeutics — Apr. 2021 – Nov. 2022
Cancer Immunology – mRNA Vaccine Development — *Shanghai, China*

– Led a three-person team that designed and screened 10+ candidate mRNA vaccine sequences using six cell-based assays, advancing three leads into three in vivo antitumor-efficacy studies for KRAS-mutant lung cancer.

– Partnered with two CROs to execute three IND-enabling studies covering efficacy, toxicology, biodistribution, and dose selection, supporting a vaccine program that progressed to Phase II clinical evaluation.

Selected Projects

HELP-E: Memory-Augmented LLM Agent for Multi-Session Reasoning [GitHub] — Mar. 2026 – May 2026
Multi-agent LLM system with graph memory and retrieval — *Dallas, TX*

– Built and Dockerized a multi-agent LLM pipeline that extracts user intent, beliefs, and behavioral states into Pydantic-validated schemas and typed graph memory; served the system through FastAPI with OpenAI-compatible vLLM backends.

– Implemented behavioral-state tracking, dense MiniLM retrieval with MMR, and Pydantic-validated structured outputs; improved primary-concern F1 from 53.3% to 73.3%.

– Designed reproducible ablations over 4,800 simulated dialogue turns and 30 multi-session profiles using LLM-as-a-judge metrics, JSON audit traces, and failure-mode analysis.

Team Lead, BioTech Club at UTD | SQL Injection Detection with DistilBERT — Jul. 2024 – Jul. 2025
NLP security classification and production ML lifecycle — *Dallas, TX*

– Led an end-to-end NLP pipeline over 100,000 curated SQL queries, implementing reproducible preprocessing, stratified data splits, baseline comparison, and DistilBERT fine-tuning to reach 0.91 F1.

– Productionized the workflow with MLflow experiment tracking, model registry, versioned promotion and rollback, and reproducible packaging; profiled CPU inference at 28 ms p50 / 46 ms p95.

Selected Publications

– **Yi Yang**, Ovidiu Daescu. "BiScale-GTR: Fragment-Aware Graph Transformers for Multi-Scale Molecular Representation Learning." *Transactions on Machine Learning Research*, 2026.

– **Yi Yang**, Arowa Yasmeen, Ovidiu Daescu. "Uncovering Structure–Performance Relationships in Organic Photovoltaics: Interpretable ML Model for Predicting the Power Conversion Efficiency." *IEEE Journal of Photovoltaics*, 2025.